

Real-Time Non-Linear Model Predictive Control and Extended Kalman Filtering for Autonomous Lunar Soft Landing

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Abstract—This paper presents a robust real-time guidance, navigation, and control (GNC) architecture for autonomous lunar lander descent. The system integrates a receding-horizon Non-linear Model Predictive Controller (NMPC) solved via interior-point methods with a 5-state Extended Kalman Filter (EKF) for dynamic state estimation under noisy telemetry. To guarantee real-time execution at 10 Hz, barrier singularities near touchdown are resolved through penalty-based ground constraint softening and strict interior iteration caps. Closed-loop flight simulations demonstrate precise touchdown at zero terminal velocity with a mean solve latency of 38.6 ms.

Index Terms—Model Predictive Control, Extended Kalman Filter, Lunar Lander, Trajectory Optimization, CasADi, Real-Time Embedded Control.

I. SYSTEM DYNAMICS & CONTROL FORMULATION

The non-linear planar dynamics of the variable-mass spacecraft are governed by:

$$\dot{x} = v_x, \quad \dot{z} = v_z \quad (1)$$

$$\dot{v}_x = \frac{T_x}{m} + d_x, \quad \dot{v}_z = \frac{T_z}{m} - g_{\text{lunar}} \quad (2)$$

$$\dot{m} = -\frac{\sqrt{T_x^2 + T_z^2}}{I_{sp}g_0} \quad (3)$$

where $g_{\text{lunar}} = 1.62 \text{ m/s}^2$, $I_{sp} = 300 \text{ s}$, and d_x represents external lateral disturbance accelerations.

II. REAL-TIME RECEDING HORIZON NMPC

The finite-horizon optimal control problem is discretized across $N = 30$ horizon steps ($T_h = 12.0 \text{ s}$). To eliminate IPOPT barrier divergence near $z = 0$, the hard ground constraint is converted into a soft quadratic penalty:

$$J_{\text{soft}} = \sum_{k=0}^N (\|x_k - x_{\text{ref}}\|_Q^2 + \|u_k\|_R^2 + \gamma \max(0, -z_k)^2) \quad (4)$$

By setting the maximum IPOPT iteration cap to $\text{iter}_{\text{max}} = 30$, worst-case latency is bounded without sacrificing tracking precision.

III. STATE ESTIMATION VIA EXTENDED KALMAN FILTER

The discrete EKF propagates non-linear dynamics and processes continuous noisy position and velocity measurements $y_k \sim \mathcal{N}(x_k, R)$:

$$\hat{x}_{k|k-1} = f(\hat{x}_{k-1|k-1}, u_{k-1}), \quad P_{k|k-1} = F_k P_{k-1|k-1} F_k^T + Q \quad (5)$$

The innovation step updates the estimate prior to feed-forward pass into the NMPC solver.

IV. RESULTS AND COMPUTATIONAL PERFORMANCE

Closed-loop simulations under wind gust disturbances (15 m/s^2 lateral pulse) proved full state convergence:

- **Touchdown Metrics:** $z = 0.2 \text{ m}$, $v_x = 0.70 \text{ m/s}$, $v_z = -0.02 \text{ m/s}$.
- **Mean Compute Latency:** 38.6 ms ($\sim 25.9 \text{ Hz}$ execution capacity).
- **Real-Time Budget Compliance:** 100% of active flight steps met the 100 ms time budget.

V. CONCLUSION

The combination of soft-constraint penalization, shift warm-starting, and EKF noise mitigation yields an execution-ready controller suitable for real-time onboard flight processors.